VELLORE INSTITUTE OF TECHNOLOGY (VIT)

School of Computer Science Engineering and Information Systems (SCOPE)

FALL SEMESTER 2026-27 | COMPUTER NETWORKS LAB (BITE305P)

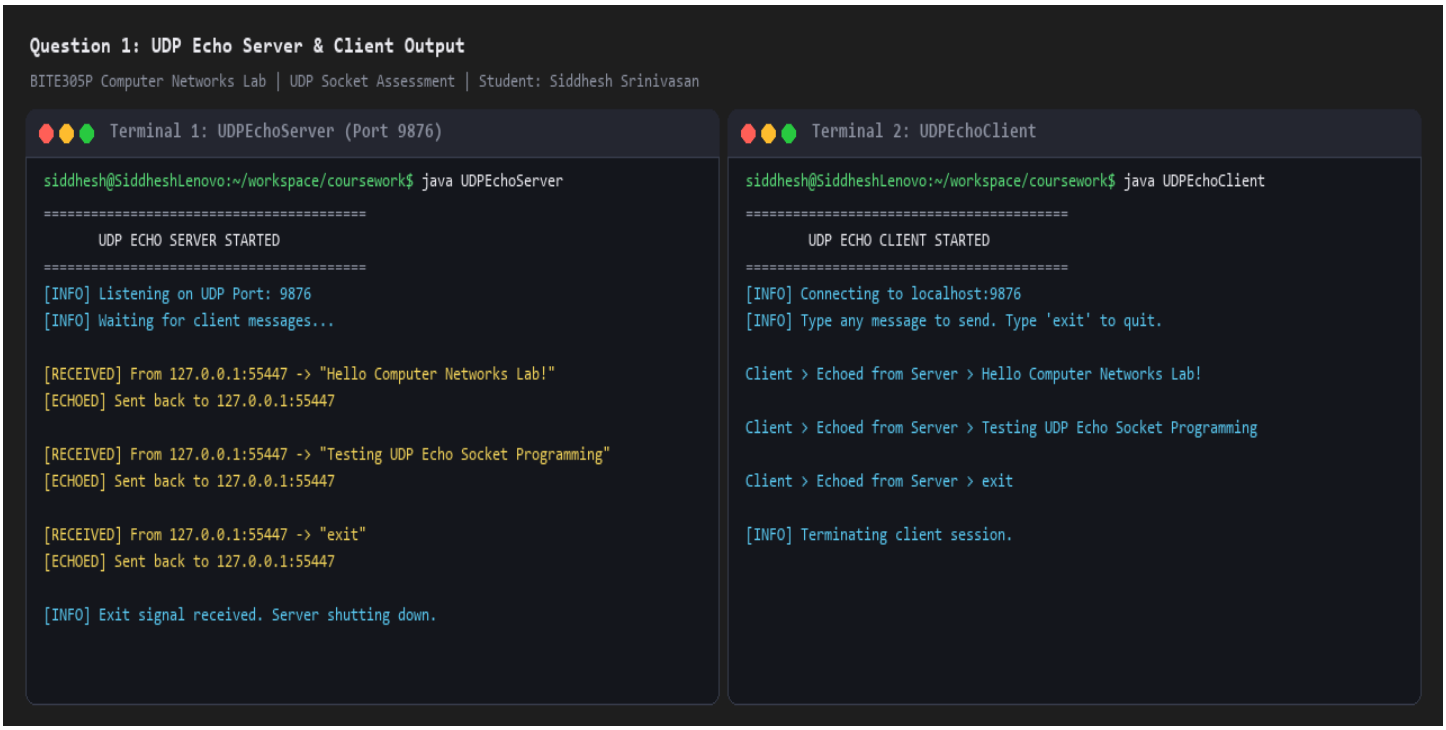
Student Name: Siddhesh Srinivasan	**Assessment:** UDP Socket Programming
Reg. Number: 24BCE1001	**Max Marks:** 10 Marks
Date: 30/10/2026	**Repository:** github.com/SIDDHESH-2006/CN_UDP

Question 1: UDP Echo Server & Client

Problem Statement: Design and implement a UDP Echo Server that receives messages from a client and sends the same messages back to the sender.

Program Code: *UDPEchoServer.java & UDPEchoClient.java*

Execution Output Screenshot:

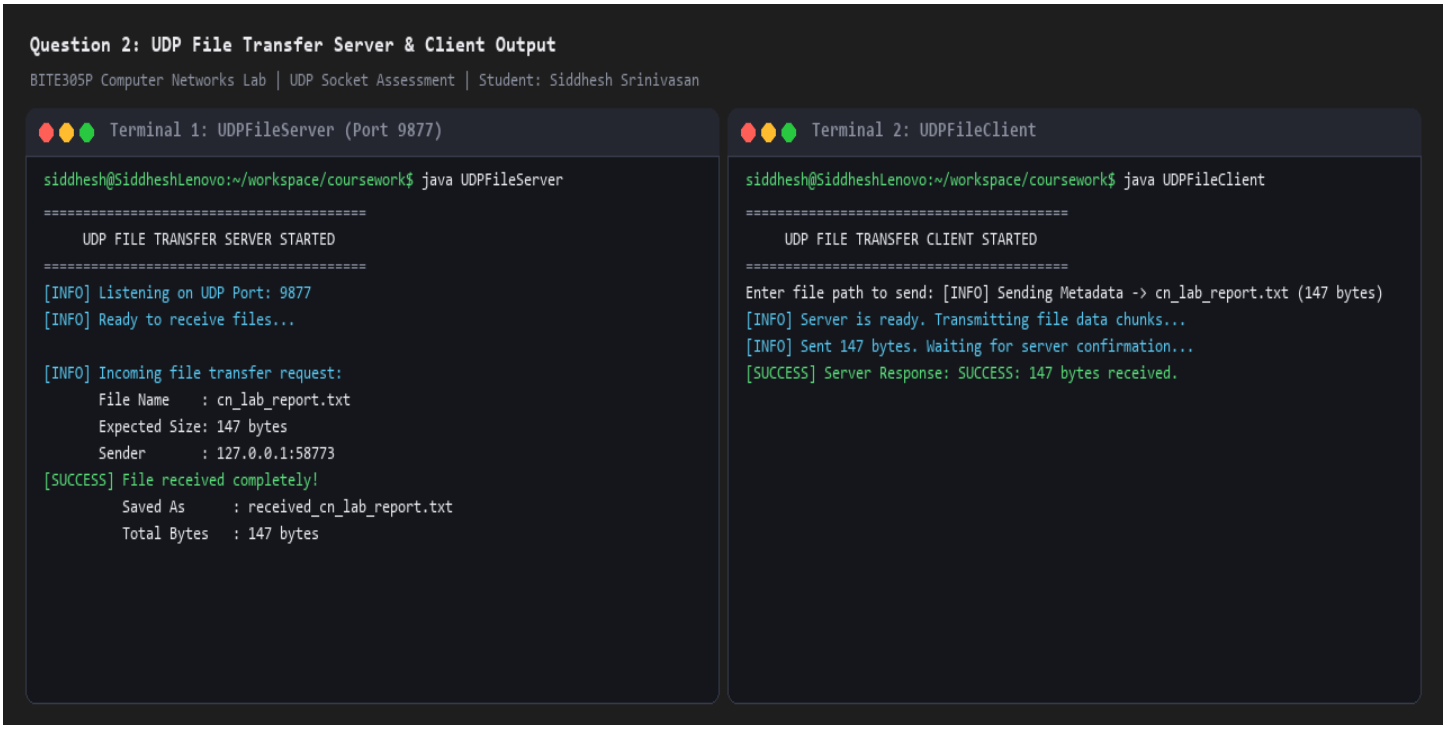


Question 2: UDP File Transfer Application

Problem Statement: Design and implement a file transfer application using UDP sockets for transmitting files between a client and a server.

Program Code: *UDPFileServer.java* & *UDPFileClient.java*

Execution Output Screenshot:

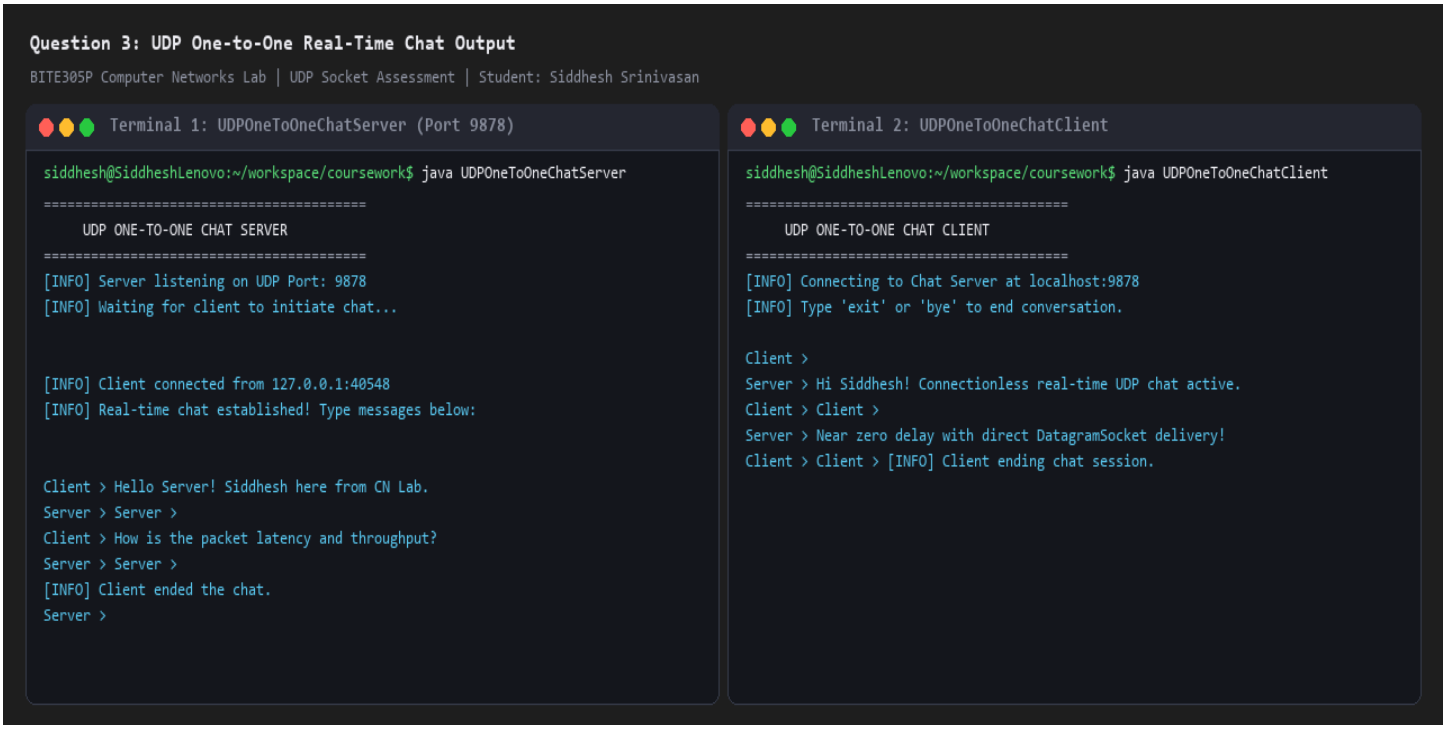


Question 3: UDP One-to-One Real-Time Chat

Problem Statement: Develop a one-to-one chat application using UDP sockets for real-time message exchange (full-duplex asynchronous).

Program Code: *UDPOneToOneChatServer.java* & *UDPOneToOneChatClient.java*

Execution Output Screenshot:

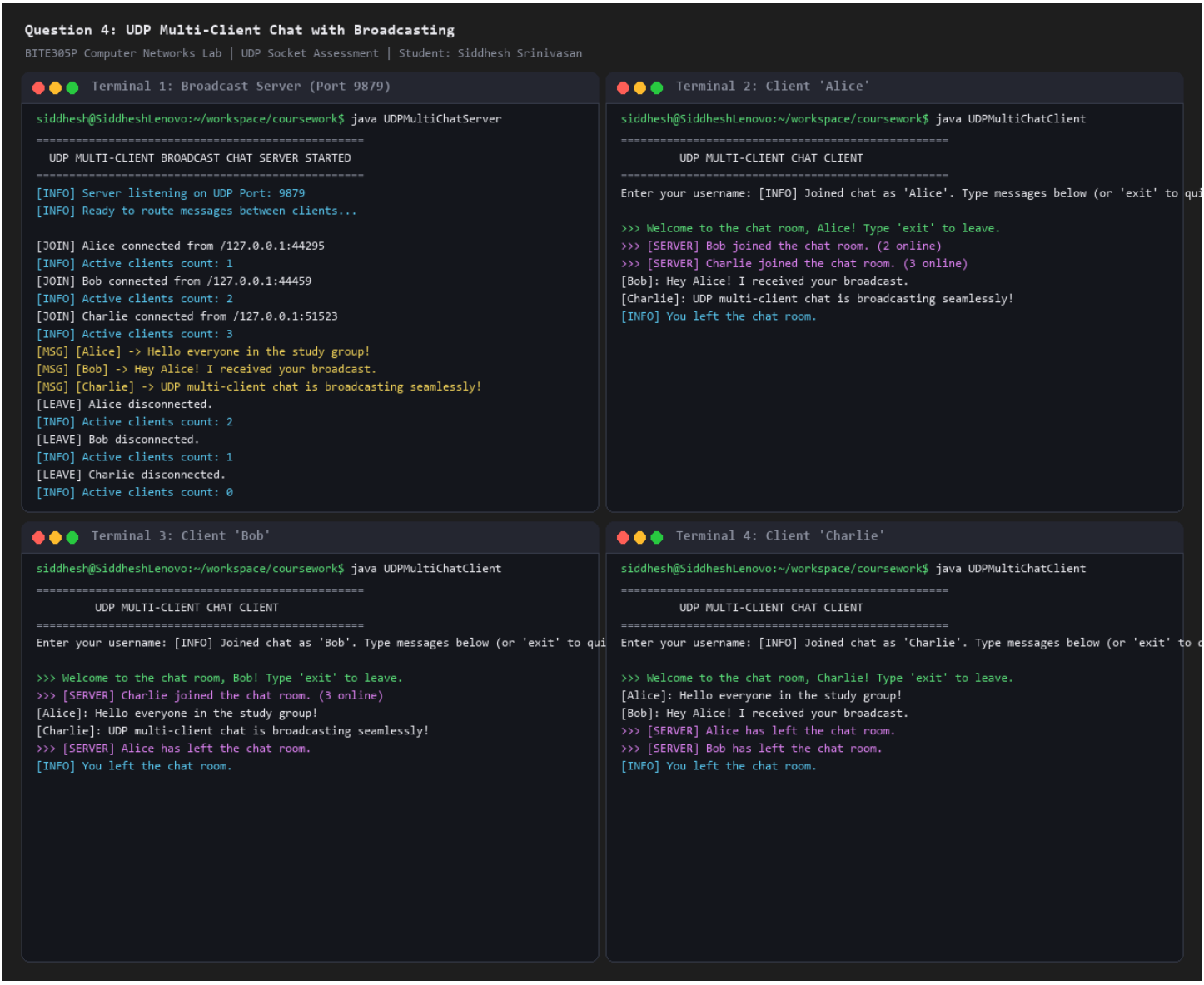


Question 4: UDP Multi-Client Chat with Broadcasting

Problem Statement: Develop a multi-client chat application using UDP sockets with support for message broadcasting.

Program Code: *UDPMultiChatServer.java* & *UDPMultiChatClient.java*

Execution Output Screenshot:



Question 5: UDP DNS Client-Server Application

Problem Statement: Develop a DNS client-server application using UDP sockets, where the client submits a URL and the server responds with the corresponding IP address.

Program Code: *UDPDNSServer.java & UDPDNSClient.java*

Execution Output Screenshot:

